

Development of a compact and controllable cold atomic source using laser ablation

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We previously developed a simple and compact cold atomic source using laser ablation. While we were successful in directly loading a strontium MOT with long lifetime from the ablated strontium vapour, the issue of strontium deposition on the glass cell limits its practical usage. In a new design, we aim to overcome this issue by reflecting the ablation beam inside the vacuum system, thus avoiding a direct line-of-sight between the ablation point and the viewport. Moreover, we will control the release of strontium vapour with a PID feedback on the 1064 nm ablation laser power. With this new design, we hope to achieve a robust and controllable compact atomic source for portable cold atomic experiments.

Motivation & Objectives

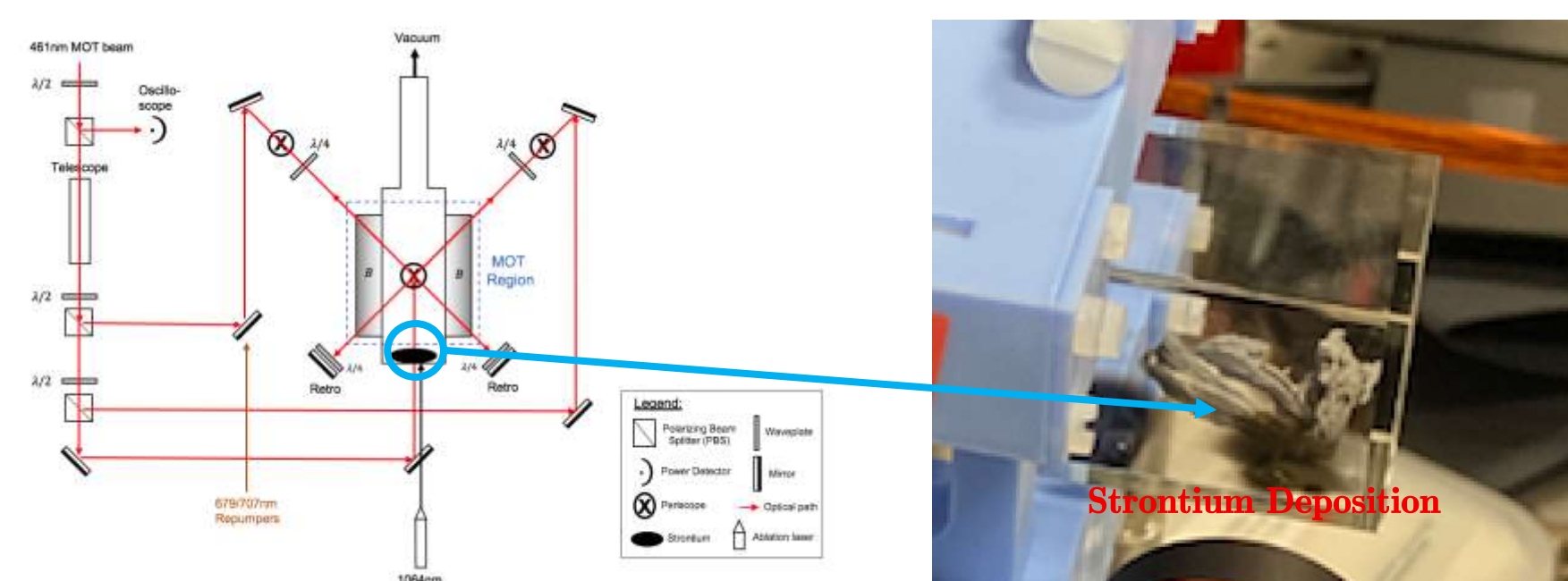


Fig.1 Previous setup and the problem

- New design should avoid the coating of strontium on the viewport for the ablation laser. This will ensure better robustness of the source
- Design and implement a PID feedback based on strontium vapour fluorescence for a controlled atomic flux and MOT loading

New Setup Design

We focus high power 1064nm laser on strontium to release the atomic vapour. The strontium vapour is detected with 461 nm fluorescence measurement.

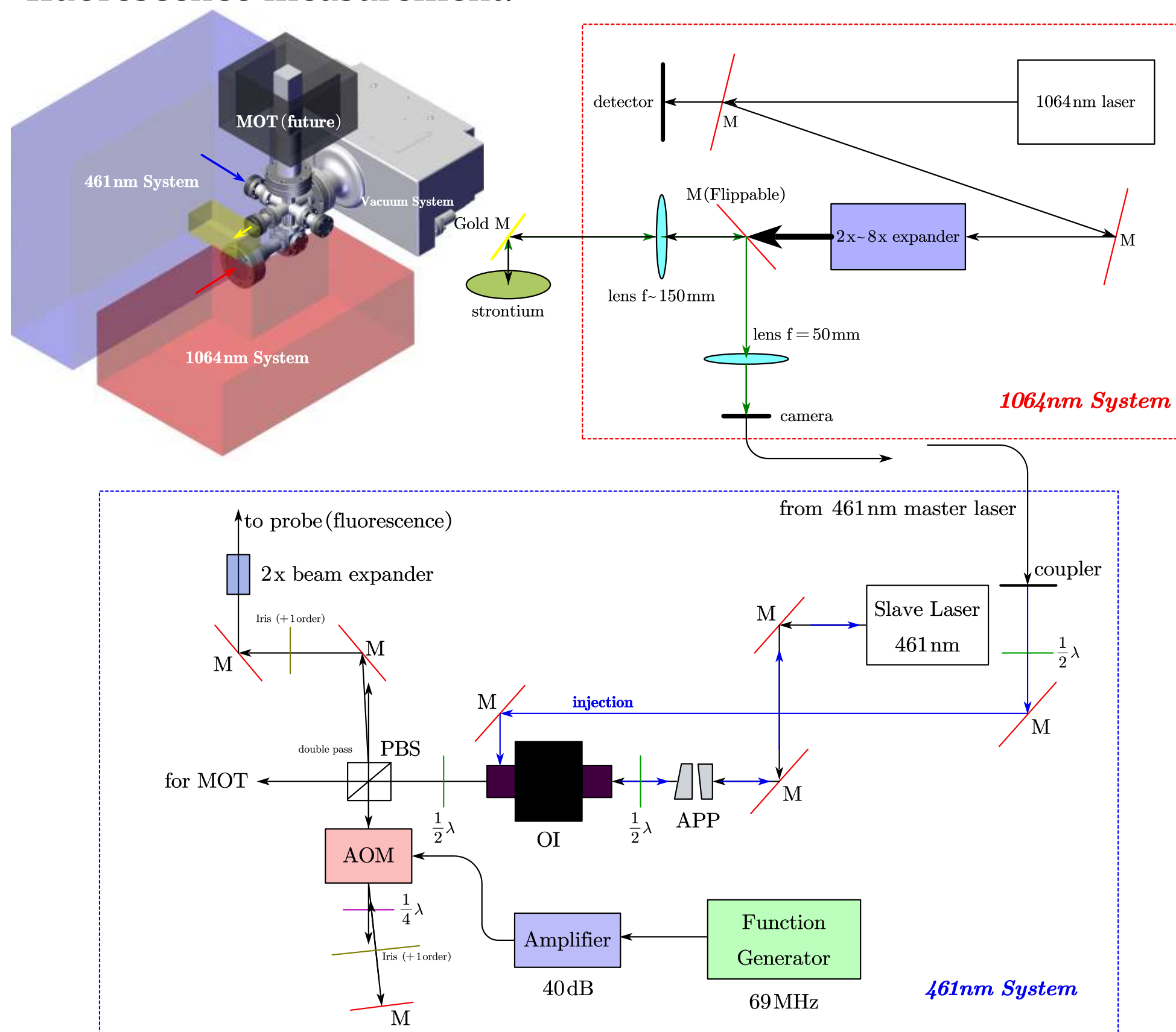


Fig.2 New setup and laser system

In our new setup, strontium is located further away from viewport. The ablation occurs by reflecting the ablation beam using a gold mirror. This avoids a direct light of sight of the ablation point and the viewport, which avoids a the deposition of strontium on the viewport.

Experimental Results

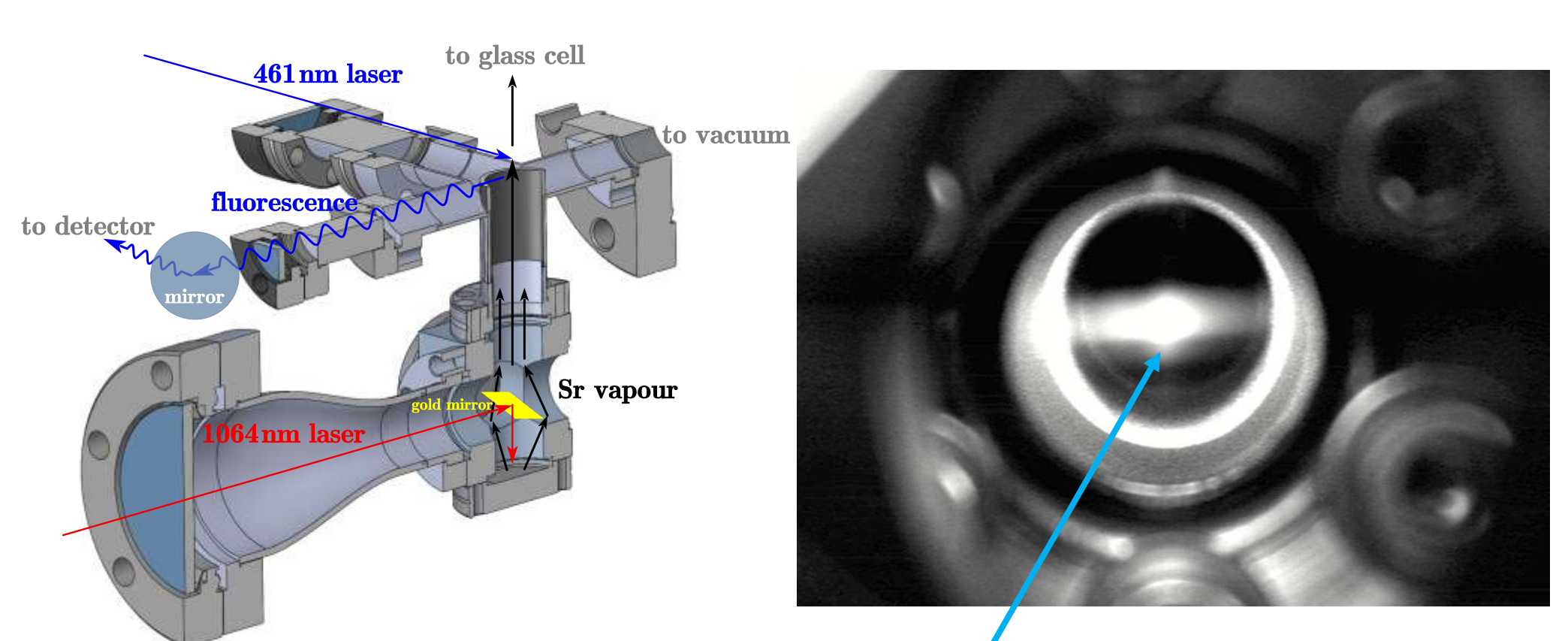


Fig.3 Fluorescence and detection

The strontium vapour released during the ablation is detected by a 461 nm fluorescence signal (photo on the right in Fig.3). For a more quantitative measurement, we use a photodetector.

With shinning pulse ablation of 15W, frequency of 2Hz and 5% duty cycle, we managed to observe fluorescence signal on an oscilloscope.

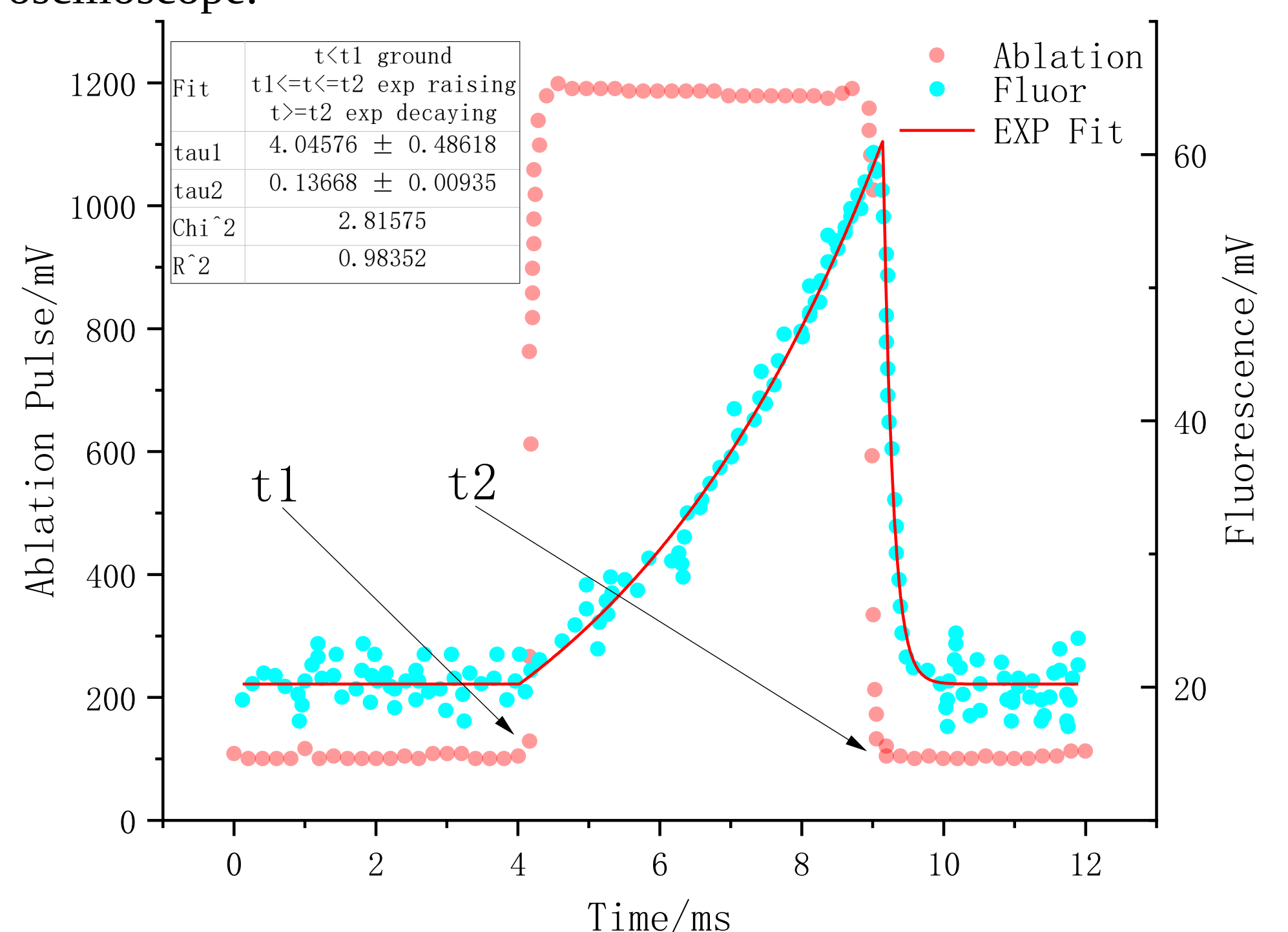


Fig.4 Signal and fit result

By fitting the detected signal to exponential functions, we find a ris-ing time of 4.05 ms and a falling time of 0.14 ms. The short falling time allows the low pressure condition to be recovered for a longer MOT lifetime.

Future work

- Optimizing 1064nm system & Enhancing the signal lifetime
- Design feedback circuit to retain constant atom number
- Construction of MOT & Trapping and cooling atoms